AgriAI — M0 Research Report

STO-Optimized LS-HGNet for Explainable Crop Disease Classification in Smart Farming

Field	Value
Project Name	AgriAI
Project Title	STO-Optimized LS-HGNet for Explainable Crop Disease Classification in Smart Farming
Full Title	Adaptive Smart Farming Framework using STO-Optimized LS-HGNet with Explainable Crop Disease Analysis
Research Stage	M0 — Research Specification
Status	✅ **LOCKED**
Date	2026-10-01 (Locked) / 2026-10-02 (Final Verification & Archive)
Team	3-member final-year project team
Target Hardware	16 GB laptop (development); standard GPU optional for final experiments

1. Abstract

Tomato is among the most widely cultivated and economically significant crops globally. Crop diseases cause substantial yield losses, and accurate, timely, and explainable automated disease diagnosis can reduce these losses and support precision agriculture. This project investigates whether a lightweight, stacked-hourglass-inspired convolutional neural network (LS-HGNet), with hyperparameters optimized using the Sooty Tern Optimization Algorithm (STOA), can deliver accurate, computationally efficient, explainable, calibrated, and robustly-evaluated tomato leaf disease classification.

The PlantVillage tomato subset is used as the primary dataset, targeting 10 classes (1 healthy + 9 diseased). The project extends concepts from prior smart-farming and bio-inspired optimization research to a new task (disease classification), with additional components including:

- **Grad-CAM explainability** — visual explanations of model predictions
- **Post-hoc temperature-scaling calibration** — reliable confidence estimates
- **Systematic robustness testing** — evaluation under 7 synthetic perturbation types
- **Flask REST API and React dashboard** — deployable inference interface

A simple CNN baseline and a Random Search optimizer are included to provide experimental controls. A project-modified stabilized/additive STOA variant is defined to address known numerical pathologies in the original STOA formulation.

This is an independent adaptation and extension project. No original research results, code, or text from the referenced source papers are reproduced.

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2. Problem Statement

Tomato leaf diseases are visually diverse and can be difficult to distinguish manually. Automated classification systems require:

1. **Sufficient classification accuracy** across multiple disease classes
2. **Computational efficiency** suitable for resource-constrained deployment
3. **Explainability** to support agricultural decision-making
4. **Calibrated confidence** to convey reliable uncertainty estimates
5. **Robustness** under realistic image degradation

Existing studies typically optimize only one or two of these properties. This project explores all five in an integrated framework.

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3. Research Gap

While deep learning methods for tomato disease classification are well-studied, the following combination has not been thoroughly investigated in the literature surveyed:

- A lightweight stacked-hourglass-inspired architecture adapted from UAV vision tasks to disease classification
- Sooty Tern Optimization used for joint hyperparameter search across learning rate, weight decay, dropout, and batch size, compared against an equal-budget Random Search under

controlled conditions
- Simultaneous evaluation of explainability (Grad-CAM), calibration (temperature scaling), and robustness (synthetic perturbation) within one experimental framework

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4. Research Question

> **Can an STO-optimized lightweight LS-HGNet architecture provide accurate, computationally efficient, explainable, calibrated, and robust tomato leaf disease classification, and does STO hyperparameter optimization yield a measurable advantage over equal-budget Random Search?**

The research question is falsifiable and will be answered through measured experimental results. No performance claims are made prior to experiments.

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5. Research Objectives

The following objectives are testable and measurable.

ID	Objective	Completion Criterion
OBJ-01	Prepare the PlantVillage tomato subset with verified class distribution, stratified split, and augmentation policy	Split CSVs created; class counts verified from downloaded data
OBJ-02	Implement a simple baseline CNN (~500K parameters)	Baseline achieves a measurable accuracy; parameter count confirmed
OBJ-03	Implement and evaluate an LS-HGNet adapted for disease classification	LS-HGNet trains and evaluates on all four metrics
OBJ-04	Implement STOA-based hyperparameter optimization	STOA completes without using test data; convergence history recorded
OBJ-05	Implement equal-budget Random Search as a control	Random Search completes with same evaluation count as STOA
OBJ-06	Evaluate all models on classification and computational metrics	Results table complete with measured values

ID	Objective	Completion Criterion
OBJ-07	Produce Grad-CAM visualizations for all trained models	Heatmaps generated and saved for at least 5 examples per class
OBJ-08	Apply and evaluate temperature scaling calibration	ECE and reliability diagrams produced before and after calibration
OBJ-09	Conduct robustness testing under 7 perturbation types	Accuracy vs. degradation curve produced
OBJ-10	Deploy a Flask REST API and React dashboard	API returns predictions with Grad-CAM overlays; dashboard displays results
OBJ-11	(Optional) ONNX export and edge device latency measurement	Model exported and latency recorded on target device

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6. Reference Research

6.1 Primary Source — LS-HGNet + STOA Applied to Smart Farming

Field	Value
Title	Smart Farming with Sooty Tern Optimization based LS-HGNet Classification Model
Authors	V. Gokula Krishnan, B. Vikranth, M. Sumithra, B. Prathusha Laxmi, B. Shyamala Gowri
Journal	International Journal of Experimental Research and Review
Volume	37 (special issue)
Pages	96–108
Year	2024
DOI	10.52756/ijerr.2024.v37spl.008
Publication Date	30 March 2024

Original application context:

- **Domain:** Precision smart farming — automated detection and counting of maize tassels in drone/UAV aerial photography
- **Preprocessing:** Morphological bottom-hat and top-hat filtering to mitigate irregular sunlight reflections and ambient shadow noise in aerial foliage
- **Network:** Lightweight Weight-Stacked Hourglass Network (LS-HGNet) with stacked hourglass modules, dilated convolutions, and inter-module skip connections
- **Optimization:** Sooty Tern Optimization Algorithm (STOA) for tuning classification hyperparameters, reporting an empirical maize tassel detection accuracy of 98.7%

> **Critical distinction:** The original paper concerns UAV-based maize tassel detection and counting. AgriAI does NOT reproduce this task. AgriAI adapts the conceptual combination of (1) a multi-stack hourglass network and (2) STOA metaheuristic tuning to a completely different application: 10-class tomato leaf disease classification on the PlantVillage dataset.

What AgriAI adopts conceptually:

- The synergistic combination of a multi-stack hourglass network parameterized for lightweight vision inference and gradient-free bio-inspired STOA metaheuristic tuning for hyperparameter search

What AgriAI does NOT adopt:

- Maize tassel counting, UAV imagery, morphological preprocessing as mandatory core pipeline, and original source weights or code

6.2 STOA Original Paper

Field	Value
Title	STOA: A bio-inspired based optimization algorithm for industrial engineering problems
Authors	Gaurav Dhiman and Amandeep Kaur
Journal	Engineering Applications of Artificial Intelligence
Volume	82
Pages	148–174
Year	2019
DOI	10.1016/j.engappai.2019.03.021

6.3 Stacked Hourglass Network

Field	Value
Title	Stacked Hourglass Networks for Human Pose Estimation
Authors	Alejandro Newell, Kaiyu Yang, Jia Deng
Year	2016
Venue	ECCV
arXiv	1603.06937

6.4 PlantVillage Dataset

Field	Value
Title	An open access repository of images on plant health to enable the development of mobile disease diagnostics
Authors	David P. Hughes, Marcel Salathé
Year	2015
arXiv	1511.08060

6.5 Additional References

Source	What we use
Howard et al. (2017), MobileNets (arXiv:1704.04861)	Depthwise separable convolution as a lightweight building block
Selvaraju et al. (2017), Grad-CAM (ICCV, DOI: 10.1109/ICCV.2017.74)	Grad-CAM technique for post-hoc explainability
Guo et al. (2017), On Calibration of Modern Neural Networks (ICML)	Temperature scaling as post-hoc calibration; ECE metric
He et al. (2016), Deep Residual Learning (CVPR)	Residual block design
Yu & Koltun (2016), Dilated Convolutions (ICLR, arXiv:1511.07122)	Dilated convolution concept

Source	What we use
Kudela (2022), Swarm and Evolutionary Computation, 74, 101133	Independent mathematical critique of STOA numerical pathologies
Kwok (2025), arXiv:2511.17556	Independent analysis of STOA origin bias and numerical divergence

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7. STOA Research Foundation

7.1 Algorithm Identity

The Sooty Tern Optimization Algorithm (STOA) is a population-based metaheuristic optimizer proposed by Gaurav Dhiman and Amandeep Kaur in 2019, published in *Engineering Applications of Artificial Intelligence*, Vol. 82, pp. 148–174 (DOI: 10.1016/j.engappai.2019.03.021).

STOA is inspired by the migration and prey-attacking behavior of sooty terns and consists of two phases:

1. **Exploration (Migration):** Agents move while avoiding collisions with neighbors and converging toward the best-known position.
2. **Exploitation (Attacking):** Agents perform spiral motion toward the best position.

7.2 Role in AgriAI

STOA is used as a gradient-free population-based hyperparameter optimization method. Its effectiveness is evaluated experimentally against a comparable Random Search baseline under an equal evaluation budget. STOA is NOT claimed to guarantee convergence to the global optimum, eliminate local minima, or be provably better than Random Search.

7.3 Original STOA Formulation (Source-Derived)

> **Verification Status: Literature-Verified, Publisher-Unverified**
>
> The documented STOA equations are consistent with independently consulted secondary literature (ResearchGate, PubMed/NIH, AIP-affiliated repositories, peer-reviewed survey papers). The original Dhiman & Kaur (2019) publisher PDF was not directly inspected during M0. Therefore, no claim of "publisher-verified" or "primary-source verified" status is made. If

> institutional/publisher access later reveals a discrepancy, the equations must be updated before publication.

Migration Phase (Exploration)

1. **Collision Avoidance:**

   $$\vec{C}_{sa}(t) = S_A \cdot \vec{P}(t)$$

   $$S_A = C_f - \left(t \cdot \frac{C_f}{T_{max}}\right)$$

   where $t \in [0, T_{max}]$ is the current iteration, T_{max} is the maximum iteration budget, and $C_f = 2.0$ decreases linearly to 0.0. Thus $S_A = 2 \cdot (1 - t / T_{max})$.

2. **Convergence Toward Best Individual:**

   $$\vec{M}*s(t) = B_s \cdot (\vec{P}*{best}(t) - \vec{P}(t))$$

   $$B_s = 0.5 \cdot \vec{r}$$

   where $\vec{r} \sim \text{Uniform}(0, 1)^D$.

3. **Combined Exploration Displacement:**

   $$\vec{D}*s(t) = \vec{C}*{sa}(t) + \vec{M}*s(t) = S_A \cdot \vec{P}(t) + B_s \cdot (\vec{P}*{best}(t) - \vec{P}(t))$$

Attacking Phase (Exploitation via 3D Spiral)

$$r = u \cdot \exp(k \cdot v)$$

$$x' = r \cdot \cos(k), \quad y' = r \cdot \sin(k), \quad z' = r \cdot k$$

where $k \in [0, 2\pi]$ is a randomly chosen angle, $u=1$ and $v=1$ define the default logarithmic spiral geometry.

Original Position Update (Dhiman & Kaur, 2019, Eq. 7–8):

$$\vec{P}(t+1) = \left(\vec{D}*s(t) \odot (x' + y' + z')\right) \odot \vec{P}*{best}(t)$$

where $\odot$ denotes element-wise (Hadamard) product.

7.4 Known Numerical Pathologies

Independent mathematical analyses (Kudela, 2022; Kwok, 2025) have demonstrated two critical numerical pathologies in the literal formulation:

1. **Exponential Magnitude Explosion:** $r = \exp(k)$ with $k = 2\pi$ evaluates to $\exp(2\pi) \approx 535.49$. Multiplying spatial displacements by factors exceeding 500 throws search candidates far outside bounded hyperparameter intervals, forcing constant boundary-clamping saturation.
2. **Structural Zero-Bias (Coordinate-System Dependency):** The multiplicative position update $\vec{P}(t+1) \propto \vec{D}*s* *odot* *vec{P}*{best}$ causes extreme collapse toward zero whenever search variables approach the origin, severely degrading search efficiency.

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8. Original STOA vs Project-Modified STOA

This distinction is mandatory for scientific integrity.

Component	Original STOA (Dhiman & Kaur, 2019)	AgriAI Project Modification
Source	Published formulation	—
Verification status	Literature-Verified, Publisher-Unverified	—
Spiral radius	$r = \exp(k)$, unbounded	$r = \exp(b \cdot k)$ with damping $b = 0.1$
Direction normalization	Not applied	Normalized to unit direction: $\vec{\Omega}_3 = [x', y', z']^T / (\|[x', y', z']\|_2 + \epsilon)$
Position update	Multiplicative Hadamard: $\vec{P}(t+1) = (\vec{D}*s* *odot (x'+y'+z')) *odot* *vec{P}*{best}$	Additive: $\vec{P}(t+1) = \vec{P}_{best}(t) + \vec{D}_s(t) \odot \vec{\Omega}_4$
$\vec{P}_{best}$ guidance	Source concept	Preserved (Option A: consistent best-guided)
P_{rand}	Not used in stabilized formulation	Not used
3D → D-dimension mapping	Not specified	Project-specific deterministic cyclic extension
Boundary handling	Not fully specified	Coordinate-wise clamping: $\vec{P}_i = \text{clip}(\vec{P}_i, lb_i, ub_i)$

Component	Original STOA (Dhiman & Kaur, 2019)	AgriAI Project Modification
Hyperparameter domain mapping	Not specified	Log-space for LR and WD; linear for dropout; discrete index for batch size

> **Attribution:** All modifications in the right column are **project-specific engineering decisions**. They are NOT attributed to the original Dhiman & Kaur STOA paper and must NOT be described as source-derived.

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9. Stabilized STOA Mathematical Specification

> **Project-Specific Modification Notice:** This is a project-specific modification inspired by the original STOA. It is not presented as the original Dhiman & Kaur STOA formulation. It must NOT be described as "primary-source verified STOA." All design choices below are engineering decisions introduced by this project for practical numerical stability during neural network hyperparameter optimization.

9.1 Search Vector

The hyperparameter search space has dimension $D = 4$:

$$\vec{P} = [\theta_1, \theta_2, \theta_3, \theta_4]^T$$

where:

- θ_1: learning rate (log-space)
- θ_2: weight decay (log-space)
- θ_3: dropout probability (linear space)
- θ_4: batch size (continuous index, mapped to discrete set)

9.2 Normalized and Damped Spiral Geometry

$$k \sim \text{Uniform}(0, 2\pi)$$

$$r = \exp(b \cdot k) \quad \text{with project damping factor } b = 0.1$$

$$x' = r \cdot \cos(k), \quad y' = r \cdot \sin(k), \quad z' = r \cdot k$$

$$\vec{\Omega}_3 = \frac{[x', y', z']^T}{|[x', y', z']|_2 + \epsilon} = \frac{[x', y', z']^T}{\sqrt{(x')^2 + (y')^2 + (z')^2} + \epsilon} = [\Omega_1, \Omega_2, \Omega_3]^T$$

where $\vec{\Omega}_3 \in \mathbb{R}^3$ is the normalized 3D direction vector and $\epsilon = 10^{-8}$.

9.3 Project-Specific 3D → D-Dimension Cyclic Extension

Because the STOA spiral produces a 3D direction vector $\vec{\Omega}_3 \in \mathbb{R}^3$ while the AgriAI hyperparameter search space has $D = 4$ dimensions, a deterministic cyclic extension is applied:

$$\vec{\Omega}_D[i] = \vec{\Omega}_3[(i-1) \bmod 3 + 1], \quad i \in {1, \dots, D}$$

For $D = 4$, this yields explicitly:

$$\vec{\Omega}_4 = [\Omega_1, \Omega_2, \Omega_3, \Omega_1]^T \in \mathbb{R}^4$$

> **This 3D → 4D cyclic extension is a project-specific engineering modification of the stabilized variant and is not attributed to the original STOA paper.**

This ensures that the element-wise perturbation $\vec{D}_s(t) \odot \vec{\Omega}_4$ is fully defined and dimensionally valid in $\mathbb{R}^4$.

9.4 Consistent Best-Guided Displacement (Option A)

Search agents are guided toward the best-so-far solution $\vec{P}_{best}$ without unintended random-agent drift:

$$\vec{C}_{sa}(t) = S_A(t) \cdot \vec{P}(t)$$

$$\vec{M}*s(t) = 0.5 \cdot \vec{r} \odot (\vec{P}*{best}(t) - \vec{P}(t))$$

$$\vec{D}*s(t) = \vec{C}*{sa}(t) + \vec{M}*s(t) = S_A(t) \cdot \vec{P}(t) + 0.5 \cdot \vec{r} \odot (\vec{P}*{best}(t) - \vec{P}(t))$$

with linearly decaying collision parameter $S_A(t) = 2 \cdot (1 - t / T_{max})$ and $\vec{r} \sim \text{Uniform}(0, 1)^4$.

9.5 Additive Position Update

$$\vec{P}(t+1) = \vec{P}_{best}(t) + \vec{D}_s(t) \odot \vec{\Omega}_4$$

9.6 Hyperparameter Domain Mapping

Hyperparameter	Search Variable	Domain	Mapping
Learning rate	$\theta_1 \in [-4.0, -2.0]$	Log-space	$\text{lr} = 10^{\theta_1}$
Weight decay	$\theta_2 \in [-6.0, -2.0]$	Log-space	$\text{wd} = 10^{\theta_2}$
Dropout	$\theta_3 \in [0.0, 0.5]$	Linear	$p_{drop} = \theta_3$
Batch size	$\theta_4 \in [0, 2]$	Continuous index	$\text{batch_size} \in {16, 32, 64}$ via nearest neighbor

9.7 Boundary Handling

$$\vec{P}_i = \text{clip}(\vec{P}_i, lb_i, ub_i)$$

9.8 Fitness Function

$$\text{fitness}(\vec{h}) = 1 - \text{macro_F1}(\vec{h}, D_{val})$$

- Minimizing $(1 - \text{macro_F1})$ is equivalent to maximizing macro F1
- Macro F1 treats all classes equally; preferred over accuracy for imbalanced datasets
- **The test set is NEVER used during fitness evaluation**

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10. LS-HGNet Architecture

10.1 Important Distinction

> The original Stacked Hourglass Network (Newell et al., 2016) was designed for **human pose estimation** (structured output). The source smart-farming paper applied a variant to **UAV-**

> **based maize tassel detection**. Our LS-HGNet adaptation applies the concept to **tomato leaf disease classification**. All implementation specifics below are our **engineering decisions** unless explicitly marked as source-derived.

10.2 Source-Derived Concepts

Concept	Source
Encoder-bottleneck-decoder hourglass structure	Newell et al., 2016
Skip connections at each resolution level	Newell et al., 2016
Stacking multiple hourglass modules	Newell et al., 2016
Residual block design	He et al., 2016 (ResNet)
Depthwise separable convolution	Howard et al., 2017 (MobileNets)
Dilated convolution concept	Yu & Koltun, 2016

10.3 Our Engineering Decisions

Decision	Choice	Rationale
Task	Classification (not detection/pose)	Task adaptation
Dilation rates	[1, 2, 4]	Engineering choice only
Multi-dilated fusion	Concatenation + 1×1 conv + residual	Engineering choice
Inter-stack skip connections	Present	Engineering decision
Classification head	GAP + Dropout + Linear	Standard practice
Input resolution (dev)	128×128	Computational feasibility
Input resolution (final)	224×224	Standard scale
Number of stacks (dev)	2	Computational feasibility
Number of stacks (final)	3	Capacity experiment
Base channels (dev)	32	Lightweight objective
Base channels (final)	64	Capacity experiment

10.4 Overall Architecture

```
Input Image (H × W × 3)
        |
    Stem Block
   Conv(3×3, stride=2) → BN → ReLU → MaxPool(2×2)
        |
   Hourglass Stack 1
   (Encoder → Bottleneck → Decoder with skip connections)
        |
  Inter-stack Skip Connection  [Engineering Decision]
        |
   Hourglass Stack 2
   (Encoder → Bottleneck → Decoder with skip connections)
        |
  [Optional Stack 3 — Final Config Only]
        |
   Global Average Pooling
        |
   Dropout (p = STOA-searched)
        |
   Linear Layer → Class Logits (10 classes)
```

10.5 Hourglass Module

Encoder: L levels (L=3 dev, L=4 final), each: LightResidualBlock → MaxPool(2×2). Channels: 32→64→128 (dev); 64→128→256 (final).

Bottleneck: MultiDilatedBlock at lowest resolution.

Decoder: L levels matching encoder. Each: Bilinear upsampling (×2) → LightResidualBlock. Skip connections from encoder concatenated before each decoder block.

Upsampling: Bilinear + 1×1 conv (chosen over transposed conv for stability and parameter efficiency).

10.6 DepthwiseSeparableConv

Property	Specification
Depthwise conv	k × k, groups=in_channels, same-size padding with dilation
Pointwise conv	1×1, groups=1

Property	Specification
Normalization	BatchNorm2d after each conv
Activation	ReLU after each BN
Default kernel	3×3
Stride	Configurable, default=1
Dilation	Configurable, default=1

10.7 LightResidualBlock

Property	Specification
Primary path	DWSConv(in, mid) → BN → ReLU → DWSConv(mid, out) → BN
Skip path	Identity if in==out and stride==1; else Conv(1×1, stride) → BN
Residual addition	Primary + Skip
Post-addition activation	ReLU
Projection condition	Applied when in_ch != out_ch OR stride != 1

10.8 MultiDilatedBlock

> **Dilation rates [1, 2, 4] are an engineering choice. They are NOT from the source smart-farming paper.**

```
Input (B, C, H, W)
     |
 [d=1 branch] [d=2 branch] [d=4 branch]  (Parallel DWSConv)
     |
 Concatenation → (B, 3C, H, W)
     |
 1×1 Conv → (B, C, H, W)
     |
 Residual Addition with Input
     |
 Output (B, C, H, W)
```

11. Classification Adaptation

11.1 Task

10-class tomato leaf disease classification on the PlantVillage dataset. This is a standard multi-class classification task where each input image is assigned to exactly one of 10 classes.

11.2 Input Pipeline

```
Raw Image → Resize → Augmentation (train only) → Normalize → Tensor
```

11.3 Classification Head

```
Final Hourglass Output (B, C, H, W)
       |
 Global Average Pooling → (B, C)
       |
 Dropout(p)
       |
 Linear(C, 10) → (B, 10) raw logits
```

No softmax inside the model. PyTorch CrossEntropyLoss applies log_softmax internally; applying softmax beforehand corrupts the loss. Temperature scaling operates on raw logits. Softmax is applied only at inference time.

11.4 Class Structure

The tomato subset of PlantVillage is commonly reported to contain 10 classes:

1. Tomato — Bacterial Spot
2. Tomato — Early Blight
3. Tomato — Late Blight
4. Tomato — Leaf Mold
5. Tomato — Septoria Leaf Spot
6. Tomato — Spider Mites (Two-Spotted Spider Mite)
7. Tomato — Target Spot
8. Tomato — Tomato Mosaic Virus
9. Tomato — Tomato Yellow Leaf Curl Virus

10. Tomato — Healthy

> **NOTE:** The number of classes (10) is consistently reported across multiple published papers and must be verified against the actual downloaded dataset during M1.

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12. Dataset Plan

12.1 Known from M0

Property	Value
Name	PlantVillage
Task subset	Tomato leaf diseases
Source	Hughes & Salathé, 2015
Access	TensorFlow Datasets / Kaggle / PlantVillage GitHub
Target classes	10 (1 healthy + 9 diseased)
Approximate total	~18,000+ images (from literature)

12.2 To Be Verified During M1

Item	Status
Exact per-class image counts	TO BE VERIFIED from downloaded files
Exact total image count	TO BE VERIFIED
Class distribution / imbalance ratio	TO BE VERIFIED
Image file formats	TO BE VERIFIED
Image dimensions (original)	TO BE VERIFIED
License terms (CC0 or other)	TO BE VERIFIED

12.3 Known Limitations

1. PlantVillage images were collected under controlled laboratory conditions with uniform backgrounds.
2. Models trained on this dataset may not generalize to field images with complex, natural backgrounds.
3. Class imbalance may exist across the 10 classes.
4. The dataset does not include images of co-occurring diseases or infection at different growth stages.
5. Lighting conditions do not reflect the full range of real-world agricultural environments.
6. The dataset contains no geographic diversity metadata.

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13. Data Preprocessing

13.1 Split Strategy

```
 PlantVillage Tomato Subset (all images)
              |
     Stratified Split (by class label)
    /              |              \
Train (70%)   Val (15%)      Test (15%)
     |
  Augmentation ONLY on Training Data
     |
  Deterministic Preprocessing on Val + Test
```

Property	Value
Train proportion	70%
Validation proportion	15%
Test proportion	15%
Strategy	Stratified by class label
Random seed	42
Split files	`data/splits/train.csv`, `data/splits/val.csv`, `data/splits/test.csv`

13.2 Leakage Prevention Rules

1. Splits are created **once** before any model training.

2. Augmentation is applied **only** to training data, never to validation or test data.
3. The test set is **never** seen during hyperparameter optimization.
4. STOA fitness is computed on **validation** data only.
5. Temperature scaling calibration parameters are fitted on **validation** data only.
6. Final reported metrics are computed on the **test** set **once** per experiment, after all hyperparameter decisions are finalized.

13.3 Augmentation Policy (Training Only)

Augmentation	Parameters
Random horizontal flip	p = 0.5
Random rotation	±30 degrees, p = 0.5
Color jitter	brightness=0.3, contrast=0.3, saturation=0.3, hue=0.1
Random resized crop	scale=(0.7, 1.0)

> **Vertical Flip Policy:** Random vertical flip is deliberately excluded from the default training pipeline because vertical inversion produces biologically unnatural upside-down leaf orientations for field-captured foliar images. It is retained strictly as an optional experimental ablation (ABL-05).

13.4 Deterministic Preprocessing (All Splits)

1. Resize to target resolution (128×128 for development; 224×224 for final)
2. Convert to RGB
3. Normalize: mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]

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14. Experimental Design

14.1 Experiment Matrix

ID	Model	Optimizer / Hyperparameters	Purpose
EXP_001	Baseline CNN	Default hyperparameters	Experimental architecture control (~500K params)
EXP_002	LS-HGNet	Default hyperparameters	Architectural baseline without hyperparameter tuning
EXP_003_original	LS-HGNet	Best from Original STOA (`original_strict`)	Evaluate source-derived STOA procedure as faithfully as possible
EXP_003_stabilized	LS-HGNet	Best from Stabilized STOA (`stabilized`)	Evaluate project-modified stabilized/additive STOA variant (Default)
EXP_004	LS-HGNet	Best from Random Search (`random_search`)	Equal-budget non-metaheuristic search control (25 dev / 100 final)

Models are NOT ranked before experiments. Results are measured, not predicted.

14.2 Baseline CNN

```
 Input (H × W × 3)
       |
Block 1: Conv(3×3, 32) → BN → ReLU → MaxPool(2×2)
       |
Block 2: Conv(3×3, 64) → BN → ReLU → MaxPool(2×2)
       |
Block 3: Conv(3×3, 128) → BN → ReLU → MaxPool(2×2)
       |
Global Average Pooling
       |
Linear(128, 10) → Logits
```

Target: ~500K parameters. Purpose: Without a baseline, the added architectural complexity of LS-HGNet cannot be justified experimentally. Baseline uses default hyperparameters; no hyperparameter optimization applied.

14.3 Three Optimizer Conditions

Condition Name	Code Identifier	Description
Original/Strict STOA	`original_strict`	Evaluates source-derived STOA procedure as faithfully as possible to empirically document its behavior and known numerical properties
Stabilized STOA	`stabilized`	Evaluates the project-modified stabilized/additive STOA variant **(Default)**
Random Search Control	`random_search`	Provides a non-metaheuristic search baseline evaluated under identical budget (25 dev / 100 final evaluations)

14.4 Random Search Budget

Property	Development	Final
STOA budget	5 agents × 5 iters = 25 evals	10 agents × 10 iters = 100 evals
Random Search budget	25 evaluations	100 evaluations
Search space	Same as STOA	Same as STOA
Fitness function	Same as STOA	Same as STOA

Equal-budget is the standard scientific protocol for optimizer comparison. If STOA uses 100 evaluations and Random Search uses 25, any difference is attributable to budget, not algorithm design.

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15. Optimization Search Space

Hyperparameter	Range	Scale	Search Variable
Learning rate	[1e-4, 1e-2]	Log	$\theta_1 \in [-4.0, -2.0]$
Weight decay	[1e-6, 1e-2]	Log	$\theta_2 \in [-6.0, -2.0]$
Dropout probability	[0.0, 0.5]	Linear	$\theta_3 \in [0.0, 0.5]$
Batch size	{16, 32, 64}	Discrete	$\theta_4 \in [0, 2]$ mapped via nearest neighbor

All 4 hyperparameters are optimized simultaneously.

Evaluation Budgets

Phase	STOA	Random Search
Development	5 agents × 5 iterations = 25 evaluations	25 evaluations
Final	10 agents × 10 iterations = 100 evaluations	100 evaluations

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16. Evaluation Metrics

16.1 Classification Metrics

Metric	Scope
Accuracy	Overall
Precision	Macro, Weighted, Per-class
Recall	Macro, Weighted, Per-class
F1 Score	Macro, Weighted, Per-class
Confusion Matrix	10×10
ROC-AUC	Per-class (one-vs-rest)
PR-AUC	Per-class

16.2 Computational Metrics

Metric	Unit
Parameter count	Integer
Trainable parameter count	Integer
Model size on disk	MB
CPU inference latency	ms/image (batch=1)
Peak GPU memory	MB (if applicable)

16.3 Optimization Metrics

Metric	Description
Best fitness	Best $(1 - \text{macro_F1})$ achieved
Number of evaluations	Integer
Optimization wall-clock time	Seconds
Convergence history	Fitness vs. evaluation number

> **All metrics must be computed from actual model runs. No metric may be manually typed or fabricated.**

---

17. Explainable AI — Grad-CAM

17.1 Purpose

Grad-CAM provides post-hoc visual explanations of CNN predictions by highlighting image regions that contributed most to the model's classification decision. This supports transparency and interpretability in agricultural decision-making contexts.

17.2 Procedure

1. Forward pass: record activations A^k at last decoder block of final hourglass stack.
2. Backward pass: compute gradient $\partial y^c / \partial A^k$.
3. Neuron importance: $\alpha^c_k = \frac{1}{Z} \sum_{i,j} \frac{\partial y^c}{\partial A^k_{ij}}$.
4. Weighted combination: $L^c = \text{ReLU}(\sum_k \alpha^c_k \cdot A^k)$.
5. Upsample to input resolution.
6. Apply colormap and overlay on original image.

17.3 Target Layer

Output of the final decoder block of the last hourglass stack, before Global Average Pooling.

17.4 Limitations

> Grad-CAM shows image regions contributing to the model's prediction. It does NOT prove that the model identified the true biological cause of the disease. Highlighted regions may reflect superficial image features (e.g., background texture, leaf shape) rather than genuine disease lesion patterns. Outputs must be interpreted with appropriate caution.

---

18. Confidence Calibration

18.1 Temperature Scaling

Post-hoc method. A learned scalar $T > 0$ divides the logits:

$$\hat{p}^k = \frac{\exp(z^k / T)}{\sum_j \exp(z^j / T)}$$

18.2 Fitting Procedure

1. Train model to convergence.
2. Minimize NLL on validation data: $T^* = \arg\min_T \text{NLL}(D_{val}; T)$.
3. Apply T^* at inference time.

The test set is NEVER used to fit T.

18.3 Metrics

Metric	When
ECE (Expected Calibration Error)	Before and after calibration
Reliability diagram	Before and after calibration
Mean raw confidence	vs. actual accuracy
Mean calibrated confidence	vs. actual accuracy

18.4 Limitations

> Calibration does not guarantee correctness; it adjusts confidence, not accuracy. Model

> confidence must NOT be equated with disease severity or treatment urgency.

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19. Robustness Testing

19.1 Perturbation Types

Perturbation	Description
Brightness increase	Pixel values multiplied by factor > 1
Brightness decrease	Pixel values multiplied by factor < 1
Contrast change	Linear contrast scaling
Gaussian noise	Zero-mean Gaussian noise, varying sigma
Gaussian blur	Varying kernel size
JPEG compression	Varying quality factor
Low-light transformation	Simulated dark/underexposed conditions

Each perturbation is applied at multiple severity levels.

19.2 Purpose

Robustness testing evaluates how model accuracy degrades under synthetic image perturbations that simulate real-world conditions (varying lighting, camera quality, compression artifacts). The output is accuracy-vs-severity degradation curves.

19.3 Robustness Claims Rule

> The model MUST NOT be described as "robust" unless measured results support that claim. This section presents accuracy-vs-severity degradation curves, not qualitative judgments.

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20. Application Architecture

20.1 System Architecture

```
 React + Vite Dashboard
         ↓ HTTP
    Flask REST API
         ↓
    AgriAI Model (PyTorch)
         ↓
Prediction + Grad-CAM Overlay + Calibrated Confidence
```

20.2 Flask REST API

Endpoint	Method	Description
`/predict`	POST	Input: leaf image. Output: JSON with class, confidence, calibrated confidence, Grad-CAM overlay (base64)
`/health`	GET	Health check

20.3 React Dashboard

- Image upload interface
- Prediction display with confidence bar
- Grad-CAM heatmap visualization
- Disease information panel

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21. Optional Extensions

The following components are **OPTIONAL** and are NOT part of core evaluated deliverables:

Extension	Status
ONNX export for edge deployment	Optional (OBJ-11)
ESP32 / microcontroller integration	Optional
Soil/temperature/humidity sensors	Optional
MQTT IoT protocol	Optional
Multimodal sensor fusion	Optional

Extension	Status
Drone / field imagery	Optional
PWA / offline functionality	Optional

These must NOT be treated as M1 requirements.

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22. Originality and Contribution

22.1 Originality Statement

This project is an independent adaptation and extension of concepts reported in prior LS-HGNet and Sooty Tern Optimization research. The original research context (UAV-based maize tassel detection and counting), dataset, task, implementation, experiments, and reported results are not reproduced as original work.

The present project independently implements and evaluates a tomato leaf disease classification system using related architectural and optimization concepts, together with additional explainability (Grad-CAM), confidence calibration (temperature scaling), robustness testing, and deployment components (Flask API, React dashboard).

All source code, experimental results, figures, and numerical values are independently produced. No text, code, or figures from the referenced source papers are copied, paraphrased without attribution, or claimed as our own findings.

22.2 Project-Specific Contributions

Contribution	Type
Tomato leaf disease classification adaptation	Task adaptation
PlantVillage dataset application	Dataset choice
Classification head (GAP + Dropout + Linear)	Engineering decision
Selected channels, stack count, dilation rates	Engineering decisions
STOA hyperparameter search space definition	Engineering decision
Project-Modified Stabilized/Additive STOA variant	Project modification

Contribution	Type
3D → 4D deterministic cyclic mapping	Project modification
Equal-budget Random Search control	Experimental design
Grad-CAM explainability integration	Project extension
Temperature scaling calibration	Project extension
Systematic robustness testing (7 perturbation types)	Project extension
Flask REST API + React dashboard	Deployment component

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23. Limitations

The following limitations must appear in all final reports and presentations:

1. PlantVillage images are from controlled conditions. Generalization to field images is not guaranteed.
2. Class imbalance may exist and must be measured and discussed.
3. Single-crop scope: trained only on tomato leaves.
4. STOA is a heuristic with no convergence guarantee to the global optimum.
5. Optimization quality depends on search space definition and evaluation budget.
6. Grad-CAM provides coarse localization and does not causally identify disease regions.
7. Calibration does not guarantee correctness; it adjusts confidence, not accuracy.
8. Experiments were conducted on a 16 GB laptop; results may differ on other hardware.
9. This is a research prototype. It is NOT a replacement for expert agricultural diagnosis.
10. No pesticide dosage recommendations are generated or implied.
11. IoT and edge deployment are optional and are not part of core evaluated deliverables.
12. **Original STOA equation verification status: Literature-Verified, Publisher-Unverified.** Direct full-text publisher PDF access to Dhiman & Kaur (2019) was not achieved during M0. This limitation is acknowledged; the project proceeds on the basis of strong multi-source literature consensus. If discrepancies are discovered upon M1 implementation (e.g., via institutional library access), the equations must be updated before publication.

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24. Research Workflow

```
M0 — Research Specification (LOCKED)
  ↓
```

```
M1 — Dataset & Environment Setup
   ↓
   Dataset Acquisition → Audit → Stratified Split
   ↓
M2 — Model Implementation
   ↓
   Baseline CNN → LS-HGNet → STOA / Random Search
   ↓
M3 — Training & Evaluation
   ↓
   EXP_001 through EXP_004 → Metrics Collection
   ↓
M4 — Analysis & Explainability
   ↓
   Grad-CAM → Calibration → Robustness Testing
   ↓
M5 — Deployment
   ↓
   Flask API → React Dashboard
   ↓
M6 — Optional Extensions
   ↓
   IoT / Edge / Drone (if pursued)
```

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25. Team Responsibilities

Member 1 — Machine Learning

- LS-HGNet architecture implementation
- Baseline CNN implementation
- Training pipeline
- Model evaluation
- Grad-CAM implementation

Member 2 — Research & Optimization

- STOA implementation (original and stabilized variants)
- Random Search implementation
- Hyperparameter search experiments
- Calibration (temperature scaling)
- Robustness testing
- Research documentation

Member 3 — Application

- Flask REST API development
- React dashboard development
- API integration with trained model
- User interface design
- Optional edge/IoT extensions

Shared Responsibilities

- Data pipeline and preprocessing
- Experiment coordination
- Results analysis and reporting
- Documentation maintenance

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26. M0 Decision Log

Resolved Research Specifications

ID	Item	Resolution
RES-01	STOA Primary Authorship	Gaurav Dhiman and Amandeep Kaur (2019), *EAAI*, 82, 148–174
RES-02	Smart Farming Source Attribution	Krishnan et al. (2024), *IJERR*, 37(spl), 96–108
RES-03	Originality and Adaptation Boundary	Source: UAV maize tassel detection; Ours: tomato disease classification
RES-04	Original vs. Stabilized STOA	Formally separated into distinct specifications
RES-05	Stabilized STOA Variant Design	Normalized Ω, damped spiral b=0.1, additive update, Option A ($\vec{P}_{best}$ throughout)
RES-06	Three Optimizer Controls	`original_strict`, `stabilized`, `random_search` with equal-budget protocol
RES-07	Data Split & Augmentation	Stratified 70/15/15; vertical flip excluded from default pipeline

ID	Item	Resolution
RES-08	Search Space & Fitness	4 hyperparameters; fitness = $1 - \text{macro_F1}$ on validation set
RES-09	Original STOA Equation Verification	**Literature-Verified, Publisher-Unverified** — Formally accepted

Ablation Studies (Planned for M2/M3)

ID	Ablation
ABL-01	MultiDilatedBlock vs. single-dilation bottleneck
ABL-02	With vs. without inter-stack skip connections
ABL-03	Stack count: 1 vs. 2 vs. 3
ABL-04	With vs. without morphological preprocessing
ABL-05	128×128 vs. 224×224 input resolution
ABL-06	Depthwise-separable vs. standard convolution

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27. Deferred Questions

M0 LOCKED Decisions

All items in Sections 1–26 of this report are **locked** and frozen for implementation. They define the research foundation, architecture boundaries, optimization strategy, experimental design, evaluation methodology, and originality boundaries.

Deferred to M1/M2 (NOT M0 Blockers)

ID	Deferred Question	Resolution Milestone
DEF-01	PlantVillage exact per-class image counts and verified class distribution	M1: Count files on disk after download
DEF-02	PlantVillage dataset licensing terms (CC0 vs. other)	M1: Inspect dataset README/LICENSE

ID	Deferred Question	Resolution Milestone
DEF-03	Hardware memory feasibility: 224×224 at batch size 32 on 16 GB RAM	M1: Memory profiling; reduce to batch 16 if needed
DEF-04	Class-weighted Cross-Entropy loss benefit on macro F1	M2: Ablation after measuring class imbalance
DEF-05	Final dropout rate selection	M1/M2: Explored within STOA search space [0.0, 0.5]
DEF-06	Optional edge deployment ONNX runtime latency	OBJ-11 (optional): Benchmark during deployment phase

These items are implementation-phase verification questions, not research specification blockers.

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28. Reproducibility Requirements

Requirement	Status
Random seeds documented (42, 123, 456 for final runs)	Set in M0
Split CSV files committed before training	Pending M1
requirements.txt / environment.yaml committed	Pending M1
Configuration YAML files committed	Committed in M0
Model weights stored outside Git	Policy set in M0 .gitignore
All results generated by code, not manually typed	Policy set in M0
Convergence plots saved as artifacts	Pending M1
Grad-CAM images saved per experiment	Pending M1

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29. References

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30. Final M0 Declaration

> **M0 RESEARCH SPECIFICATION — LOCKED**
>
> The research foundation, architecture boundaries, optimization strategy, experimental design, evaluation methodology, and originality boundaries are frozen for implementation.
>
> The original STOA formulation is classified as **Literature-Verified, Publisher-Unverified**. This limitation is explicitly documented and does not block implementation.

> The next development stage is **M1 — Dataset & Environment Setup**.

End of M0 Research Report — AgriAI — Locked 2026-10-01 — Archived 2026-10-02